

逆元练习

人员

周子一、陈洛冉、谢亚锴、司云心、杨瑾硕、秦显森、于子珈 到课, 李子瀚 线上

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4430>

王向东老师周日十点半C++逆元									
刷新									
#	用户名	姓名	编程分	时间	A	B	C	D	E
1	chenluoran	陈洛冉	500	2595	100	100	100	100	100
2	qinxiansen	秦显森	500	2621	100	100	100	100	100
3	yangjinshuo	杨瑾硕	400	1171	100	100	100	100	
4	yuzijia1	于子珈	400	1493	100	100	100	100	
5	lizihan	李子瀚	400	1607	100	100	100	100	
6	zhouziyi	周子一	400	1720	100	100	100	100	
7	xieyakai	谢亚锴	400	2231	100	100	100	100	
8	niutongze	牛同泽	366	994	100	66	100	100	
9	siyunxin	司云心	356	51	100	66	100	90	

本周作业

<https://cppoj.kids123code.com/contest/4523> (课上讲了 A ~ B 题, 课后作业是 C 题必做)

课堂表现

同学们课上听讲都很认真, 就是做题的时候细节把握不好, 有时候忘了 +mod, 有时候忘了 %mod, 以后要细心注意。

课堂内容

Colorful Blocks

枚举 i 组相邻的颜色一样, 则方案数为: $C(n-1, i) * m * qmod(m-1, n-1-i)$

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2e5 + 5;
const int mod = 998244353;
int fac[maxn], inv_fac[maxn];

int qmod(int a, int k) {
```

```

int res = 1;
while (k) {
    if (k&1) res = res*a % mod;
    a = a*a % mod;
    k >>= 1;
}
return res;
}

int C(int n, int m) {
    return fac[n] * inv_fac[m] % mod * inv_fac[n-m] % mod;
}

signed main()
{
    int n, m, K; cin >> n >> m >> K;

    fac[0] = 1;
    for (int i = 1; i <= n; ++i) fac[i] = fac[i-1]*i % mod;
    inv_fac[n] = qmod(fac[n], mod-2);
    for (int i = n-1; i >= 0; --i) inv_fac[i] = inv_fac[i+1]*(i+1) % mod;

    int res = 0;
    for (int i = 0; i <= K; ++i) {
        int sum = C(n-1, i) * m % mod * qmod(m-1, n-1-i) % mod;
        res = (res + sum) % mod;
    }
    cout << res << endl;
    return 0;
}

```

Random Swaps of Balls

$f[i][1]$: i 轮后处在 1 位置的概率

$f[i][2]$: i 轮后处在后面位置的概率

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e5 + 5;
const int mod = 998244353;
int f[maxn][3];

int qmod(int a, int k) {
    int res = 1;
    while (k) {
        if (k&1) res = res*a % mod;
        a = a*a % mod;
    }
}

```

```

        k >>= 1;
    }
    return res;
}

signed main()
{
    int n, K; cin >> n >> K;
    int p1 = (n*n - 2*(n-1)) % mod, p2 = 2, v = qmod(n*n%mod, mod-2);
    p1 = p1*v % mod, p2 = p2*v % mod;

    f[0][1] = 1, f[0][2] = 0;
    for (int i = 1; i <= K; ++i) {
        f[i][1] = f[i-1][1]*p1 + f[i-1][2]*p2%mod*(n-1);
        f[i][2] = f[i-1][2]*p1 + f[i-1][1]*p2 + f[i-1][2]*p2%mod*(n-2);
        f[i][1] %= mod, f[i][2] %= mod;
    }

    int res = f[K][1] + f[K][2] * (n+2) % mod * (n-1) % mod * qmod(2, mod-2);
    res %= mod;

    cout << res << endl;
    return 0;
}

```

Writing a Numeral

按照题意进行模拟即可

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e5 + 5;
const int mod = 998244353;

int qmod(int a, int k) {
    int res = 1;
    while (k) {
        if (k&1) res = res*a % mod;
        a = a*a % mod;
        k >>= 1;
    }
    return res;
}

signed main()
{
    int Q; cin >> Q;

```

```
string s = " 1";
int id = 1, n = 1, val = 1;

while (Q -- ) {
    int op; cin >> op;
    if (op == 1) {
        int x; cin >> x;
        s += (x+'0'); ++n;
        val = (val*10 + x) % mod;
    } else if (op == 2) {
        int x = s[id]-'0';
        val = (val - x*qmod(10,n-id)%mod + mod) % mod;
        ++id;
    } else cout << val << endl;
}
return 0;
}
```